

Model Calibration: Accuracy, ECE, and Mean Confidence
(GSM8K test, greedy decoding, FP32 logprobs)

Optimizer	Obj.	Prec.	Accuracy	Parsable	ECE	Mean Conf.
AdamW	GRPO	FP32	0.7%	1.5%	0.3103	0.0000
AdamW	GRPO	BF16	46.3%	70.8%	0.3424	0.9966
AdamW	GRPO	Mixed	61.3%	95.4%	0.3558	0.9989
Muon	GRPO	FP32	48.9%	79.4%	0.3507	0.9664
Muon	GRPO	BF16	9.2%	14.0%	0.3385	0.9979
Muon	GRPO	Mixed	59.2%	93.4%	0.3604	0.9939
AdamW	SFT	FP32	49.7%	92.6%	0.4579	0.9939
AdamW	SFT	BF16	49.7%	89.4%	0.4247	0.9811
AdamW	SFT	Mixed	50.6%	98.9%	0.4833	0.9944
Muon	SFT	FP32	50.8%	93.2%	0.4483	0.9932
Muon	SFT	BF16	5.0%	7.9%	0.3634	0.9980
Muon	SFT	Mixed	51.2%	99.2%	0.4783	0.9931